Name: _____

BME 364
MECHANISMS OF ALLERGIC DISEASE
Lab Manual

Group I

ZO-1 immunofluorescence and confocal imaging

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Objective: To visualise the tight-junction protein ZO-1 in Caco-2 monolayers exposed to polysorbate 80 (P80; 0.02, 0.1 and 0.5 % v/v) and to quantify ZO-1 per cell as the ZO-1/DAPI ratio from confocal z-stacks.

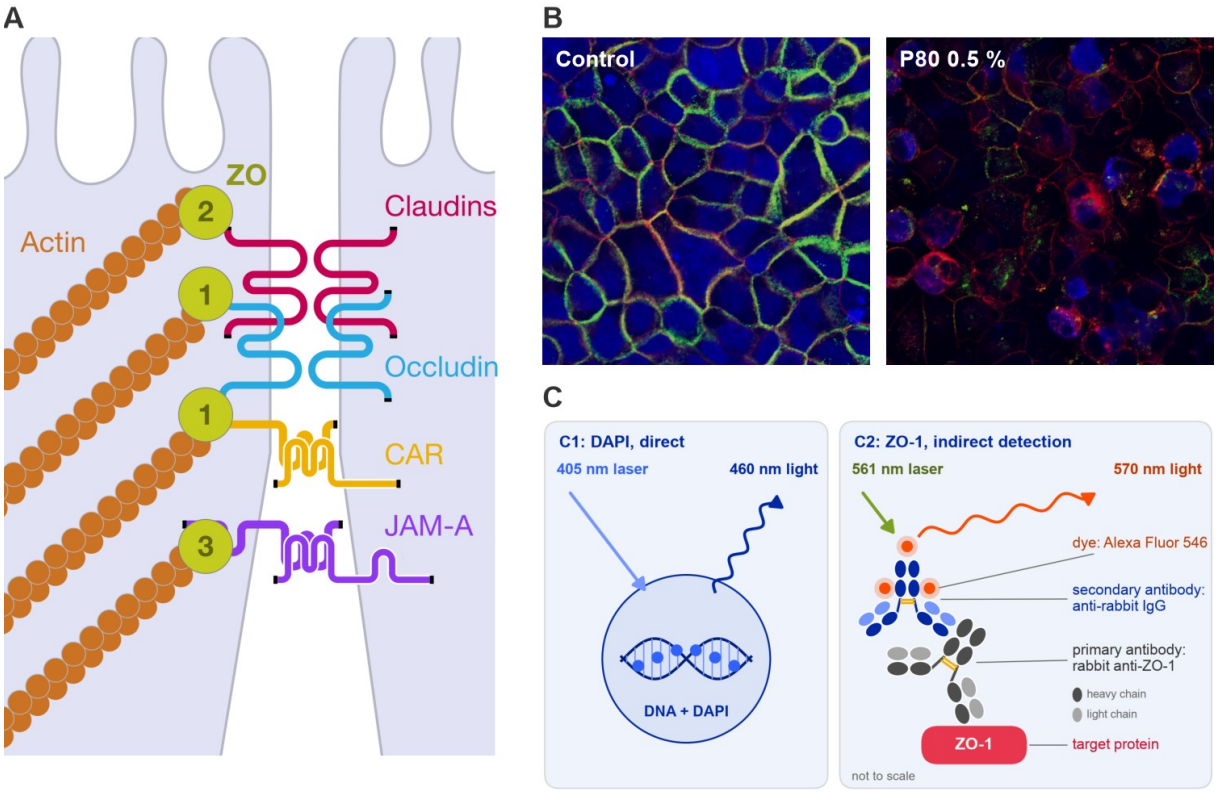


Figure 1. (A) The tight junction between two epithelial cells; Zeyneloglu, Babayev et al., FEBS Lett 2025, Fig. 2B, doi:10.1002/1873-3468.70113. (B) Real Caco-2 fields from an earlier experiment, stained for occludin and claudin-1 with DAPI (blue), maximum projection. (C) How DAPI and the ZO-1 antibodies give two separate channels (not to scale).

Background

Tight junctions and ZO-1

Tight junctions seal the space between neighbouring epithelial cells. Transmembrane proteins (claudins, occludin, CAR and JAM-A) close the gap between the cells. Inside the cell, the zonula occludens proteins ZO-1, ZO-2 and ZO-3 bind these proteins and link them to the actin cytoskeleton (Figure 1A). ZO-1 is therefore the scaffold of every tight junction. In an intact monolayer the ZO-1 staining outlines every cell, and the cells form a closed honeycomb.

What P80 does to the barrier

Polysorbate 80 (P80) is a detergent that is used as an emulsifier in food. It loosens the tight junctions: the ZO-1 signal falls and the honeycomb breaks up into fragments (Figure 1B). At high concentrations the cells die. In the MTT assay almost no cells survive from 0.25 % P80 on, so the changes at 0.5 % include cell death.

How immunofluorescence makes ZO-1 visible

ZO-1 is detected by indirect immunofluorescence (Figure 1C). The cells are fixed with paraformaldehyde so that every protein stays in place. They are permeabilised

with Triton X-100 and SDS so that the antibodies can enter the cells, and blocked with BSA so that the antibodies do not stick to other sites. A primary antibody from rabbit binds ZO-1. A secondary antibody against rabbit IgG, labelled with the dye Alexa Fluor 546, then binds the primary antibody. Several secondary antibodies bind each primary antibody, which makes the signal brighter. DAPI binds DNA and stains every nucleus directly.

Each dye is excited by one laser and emits light of a longer wavelength. DAPI is excited at 405 nm and emits blue light around 460 nm. Alexa Fluor 546 is excited at 561 nm and emits orange light around 570 nm. The microscope records the two dyes as two separate channels: channel 1 (C1) is DAPI and channel 2 (C2) is ZO-1.

Confocal imaging and the maximum projection
A confocal microscope has a pinhole that blocks the light from above and below the focal plane, so every image is a thin optical section. A monolayer is not flat: the cells differ in height, and the ZO-1 belt lies at a different height in different cells. A single section is therefore sharp only where it cuts the belt, and an intact barrier would look broken. We record a z-stack of several sections and combine them into a maximum intensity projection, which keeps the brightest value along z for every pixel. Every junction then appears at its best focus. The same sections are used for every field.

Intensities can only be compared when every well is imaged with the same laser power, gain, pinhole and zoom, and when no pixel is saturated. The fields are chosen at random on the DAPI channel, before anyone looks at ZO-1.

The read-out: the ZO-1/DAPI ratio
A P80-treated field has fewer cells and gaps between them, so the ZO-1 mean of the whole field falls even if every remaining cell keeps its junctions. We therefore divide the ZO-1 mean of the whole field by its DAPI mean. DAPI measures nuclear DNA, and so the number of cells, which makes the ratio the amount of ZO-1 per cell. Gaps lower ZO-1 and DAPI together, so their effect cancels. Both means are measured raw, without background subtraction or contrast changes. The method needs no segmentation and no setting that has to be tuned for each image.

The ratio assumes that every nucleus gives the same DAPI signal. Dying cells have condensed nuclei that are brighter, so at 0.5 % P80 the ratio has to be read with care.

What counts as a replicate
All wells come from one experiment, so they are technical replicates. The fields of one well share the seeding, the P80 dose, the staining and the imaging, so they are not independent of each other. The wells are the replicates (n = 4 per condition): average the five fields of each well, or keep all fields in a mixed model with a random intercept for each well. Counting every field as a replicate makes the evidence look stronger than it is; this mistake is called pseudoreplication.

Reagents

- **Caco-2 cells** and complete culture medium (DMEM + 10 % FBS + 1 % Pen/Strep).
- **P80 (Polysorbate 80 / Tween 80):** sterile, cell-culture grade.
- **1X PBS**, sterile.
- **4 % paraformaldehyde (PFA)** in PBS: toxic, work in the fume hood.
- **Permeabilisation buffer:** 0.1 % Triton X-100 + 0.02 % SDS in PBS.
- **Blocking buffer:** 3 % BSA in PBS.
- **Primary antibody:** rabbit anti-ZO-1 (e.g. Invitrogen 61-7300), 1:100 in blocking buffer.
- **Secondary antibody:** goat anti-rabbit IgG, Alexa Fluor 546, 1:500 in blocking buffer.
- **DAPI:** 1 µg/mL in PBS (nuclei).

Equipment

- Sterile **24-well glass-bottom plate** (#1.5 cover glass, black walls).
- Humidified CO_2 incubator (37°C, 5 % CO_2), aluminium foil, fume hood.
- **Confocal microscope** with 405 nm and 561 nm lasers and a 63× oil objective.
- Fiji with the macro **fiji/measure_zo1.ijm** and R (RStudio) with **05_if_analysis.Rmd** (folder R/mac_linux or R/windows) (course folder).

Solutions

P80 Treatment Solutions (0.02, 0.1 and 0.5 % v/v)

For 10 mL of 1 % (v/v) P80 stock solution:

1. In a sterile laminar flow hood, add 9.9 mL of sterile complete culture medium (warmed to 37°C) to a 15 mL sterile conical tube.
2. Add 100 µL of sterile P80 to the medium. (Calculation: 1 % of 10 mL = 0.1 mL = 100 µL.)
3. Vortex vigorously until the P80 is completely dissolved.

Treatment solutions, 10 mL each, in complete culture medium (37°C):

1. **0.5 %:** 5 mL of the 1 % stock + 5 mL medium (1:2).
2. **0.1 %:** 1 mL of the 1 % stock + 9 mL medium (1:10).
3. **0.02 %:** 2 mL of the 0.1 % solution + 8 mL medium (1:5).
4. Vortex well. Prepare fresh on the day of treatment; the control receives complete medium only (0 % P80).

Antibody and DAPI solutions (prepare fresh, keep on ice and dark)

1. **Primary (1:100):** 35 µL anti-ZO-1 + 3465 µL blocking buffer = 3.5 mL (16 wells × 200 µL + spare).
2. **Secondary (1:500):** 7 µL Alexa Fluor 546 antibody + 3493 µL blocking buffer = 3.5 mL (16 wells × 200 µL + spare).
3. **DAPI (1 µg/mL):** 6 µL of 1 mg/mL DAPI stock + 6 mL PBS (16 wells × 300 µL + spare).

Check the dilutions against the datasheet of the antibody lot you use.

Plate map

24-well glass-bottom plate. Four wells per condition are the replicates (n = 4). Columns 5 and 6 stay empty.

	1	2	3	4	5	6
A	CTRL	CTRL	CTRL	CTRL		
B	0.02 %	0.02 %	0.02 %	0.02 %		
C	0.1 %	0.1 %	0.1 %	0.1 %		
D	0.5 %	0.5 %	0.5 %	0.5 %		

CTRL (A1–A4): Cells + complete medium (0 % P80). Full staining.
0.02 % (B1–B4), 0.1 % (C1–C4), 0.5 % (D1–D4): Cells + P80 at that concentration. Full staining.

Steps

Cell Seeding and Treatment

□ 1. Harvest and count Caco-2 cells as described in the MTT manual (Harvest Cells to Calculate Cell Concentration).

□ 2. Seed 1.2×10^5 cells in 500 µL complete DMEM into wells A1–A4, B1–B4, C1–C4 and D1–D4 (C_2 = 240,000 cells/mL; 16 wells × 500 µL = 8 mL, prepare 10 mL).

□ 3. Grow at 37°C, 5 % CO_2 until the monolayer is confluent; change the medium every 2 days.

□ 4. Aspirate the medium. Add 500 µL of 0.02 % P80 to B1–B4, 0.1 % P80 to C1–C4, 0.5 % P80 to D1–D4 and complete medium to A1–A4.

□ 5. Incubate for 24 hours at 37°C, 5 % CO_2 (the same exposure as the 24 h TEER reading). Fix directly afterwards.

Fixation

□ 6. Aspirate the medium and wash twice with 500 µL PBS. Pipette slowly against the well wall: P80-treated monolayers detach easily, at 0.5 % most cells are dead.

□ 7. Add 300 µL of 4 % PFA per well (fume hood) and fix for 15 minutes at room temperature.

□ 8. Wash three times with 500 µL PBS, 5 minutes each. (Pause point: the sealed plate can be kept in PBS at 4°C for up to one week.)

Permeabilisation and Blocking

□ 9. Add 300 µL of permeabilisation buffer (0.1 % Triton X-100 + 0.02 % SDS) for 10 minutes at room temperature.

□ 10. Wash three times with 500 µL PBS.

□ 11. Add 300 µL of blocking buffer for 1 hour at room temperature.

Antibody and DAPI Staining

□ 12. Aspirate the blocking buffer. Add 200 µL of primary antibody to all 16 wells.

□ 13. Incubate overnight at 4°C in a sealed, humidified box.

□ 14. Wash three times with 500 µL washing buffer, 5 minutes each.

□ 15. Add 200 µL of secondary antibody to all 16 wells. Wrap the plate in aluminium foil and incubate for 1 hour at room temperature. From now on, keep the plate dark.

□ 16. Wash three times with 500 µL washing buffer, 5 minutes each.

□ 17. Add 300 µL of DAPI for 10 minutes at room temperature.

□ 18. Wash twice with washing buffer and leave 500 µL PBS in each well. Store at 4°C in the dark and image within 3 days.

Confocal Imaging

□ 19. Use the 63× oil objective with a pixel size of about 0.1 µm (e.g. 900 × 900 pixels = 90 × 90 µm field).

□ 20. Record two channels in this order: **C1 = DAPI** (405 nm) and **C2 = ZO-1** (561 nm). The macro expects this order.

□ 21. Set laser power, gain, offset and pinhole (1 AU) on a control well, where ZO-1 is brightest. Use the range indicator (HiLo LUT): no saturated pixels in **either** channel: the ZO-1/DAPI ratio needs real DAPI values too.

□ 22. Keep these settings **identical for every well**. Never use auto-adjust between wells.

□ 23. Record a z-stack of 5 slices, 0.5 µm apart, that includes the apical junction plane.

□ 24. Choose 5 random, non-overlapping fields per well using the **DAPI channel only**, before you look at ZO-1.

□ 25. Save every field as a **16-bit, 2-channel TIFF z-stack** (.tif). Check one file in Fiji (Image › Show Info): 2 channels, 5 slices, pixel size ~0.1 µm. If the pixel size is missing, enter it under Image › Properties. Never save as JPEG or as an RGB snapshot.

□ 26. While imaging, fill in **image_metadata.csv**: one row per image, written immediately (see the table below).

file	plate_well	condition	p80_pct	well	field
zo1_001.tif	A1	ctrl	0	well1	1
zo1_041.tif	C1	P80-0.1	0.1	well1	1
zo1_061.tif	D1	P80-0.5	0.5	well1	1

Data analysis

Measurement in Fiji

□ 27. Put all .tif images of the experiment in one folder (two channels, all z-slices). The macro reads only files ending in .tif.

□ 28. Test the macro on one control and one P80 image first: DAPI and ZO-1 not saturated. Then keep the acquisition settings fixed.

□ 29. Run **fiji/measure_zo1.ijm** (Plugins › Macros › Run…) and select the folder. It writes **results.csv** with one row per field: Label, DAPI_Mean (whole field, raw), and ZO1_Mean (whole field, raw).

□ 30. To do one field manually, follow the "Fiji manually" slides of the Confocal Part II workshop: they show every menu click.

Statistics in R

□ 31. Copy results.csv to data/fiji_results_zo1.csv and your metadata table to data/image_metadata.csv. Knit **05_if_analysis.Rmd** (folder R/mac_linux on macOS/Linux, R/windows on Windows) in RStudio: it joins the two tables by file name.

□ 32. **ZO-1/DAPI ratio** (main ZO-1 read-out) for every field:
ZO-1/DAPI = ZO1_Mean / DAPI_Mean

□ 33. Average the 5 fields of each well. The **well means are your replicates (n = 4 per condition)**; fields are not independent.

□ 34. Compare every P80 concentration with the control: one-way ANOVA on the well means followed by Dunnett's test, or keep all fields in a linear mixed model with a random intercept per well (Dunnett contrasts).

□ 35. At 0.5 % P80 the MTT shows no living cells: read ZO-1 there as the result of cell death, not of a junction-specific effect.

□ 36. Plot a SuperPlot: all fields in the background, well means on top.

□ 37. Report the mean difference with its 95 % CI, the exact p-value, the test and n = 4 wells from one experiment. Show representative images with the same display range for all conditions.

References

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